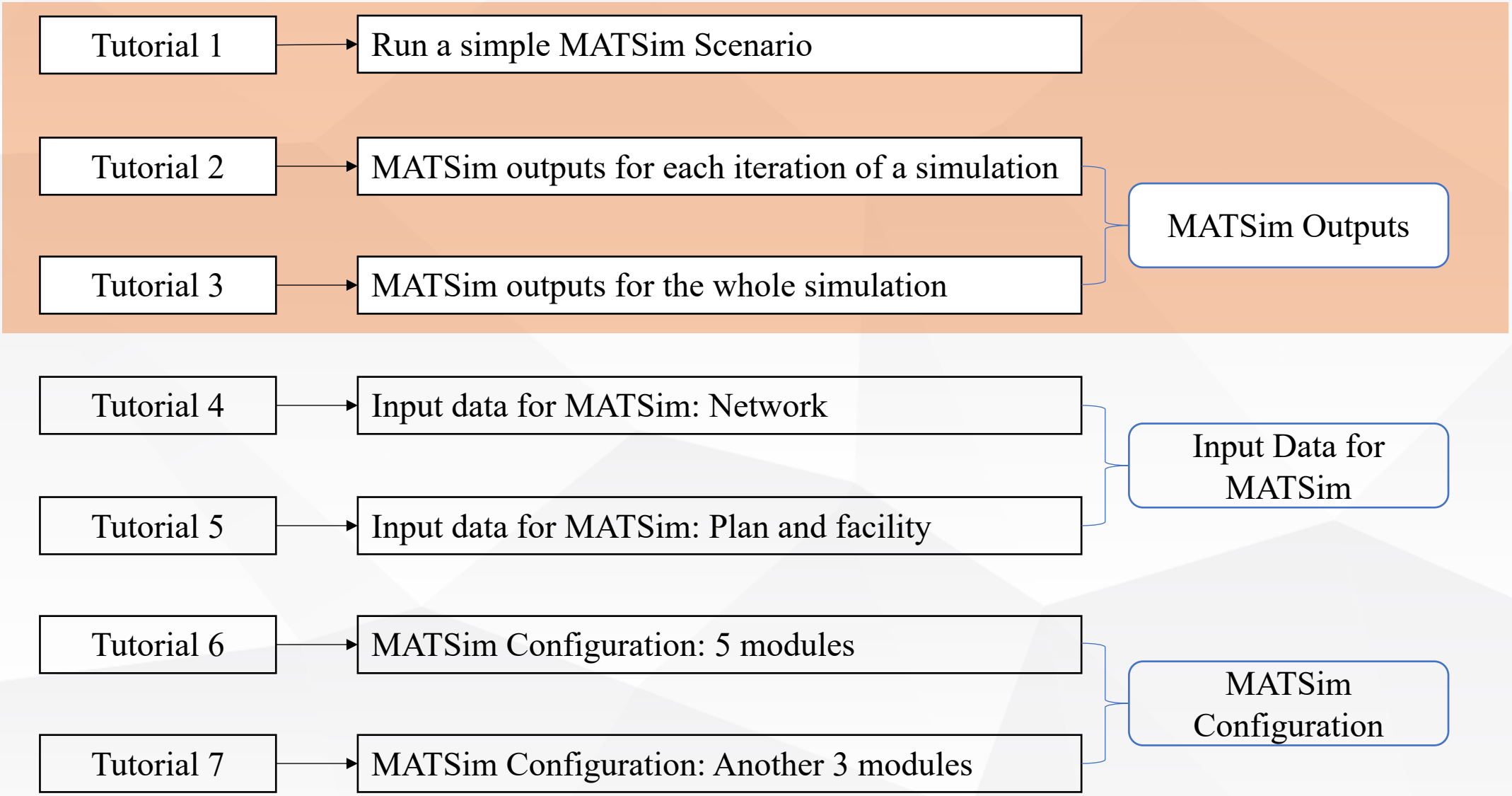
LSGI3213: Intelligent Transportation Systems

Discussion (A)

Subject Leader: Chengxiang (Tony) ZHUGE (chengxiang.zhuge@polyu.edu.hk)

Tutor: Ruichen MA (ruichen.ma@connect.polyu.hk)

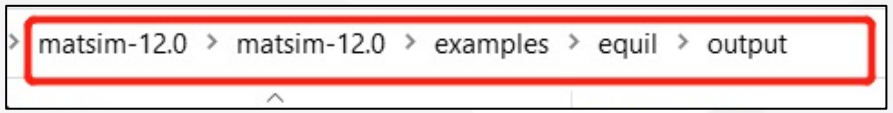
Review of Tutorials

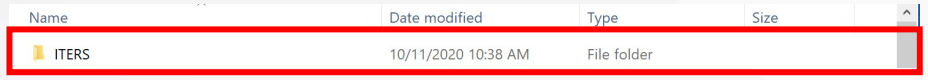


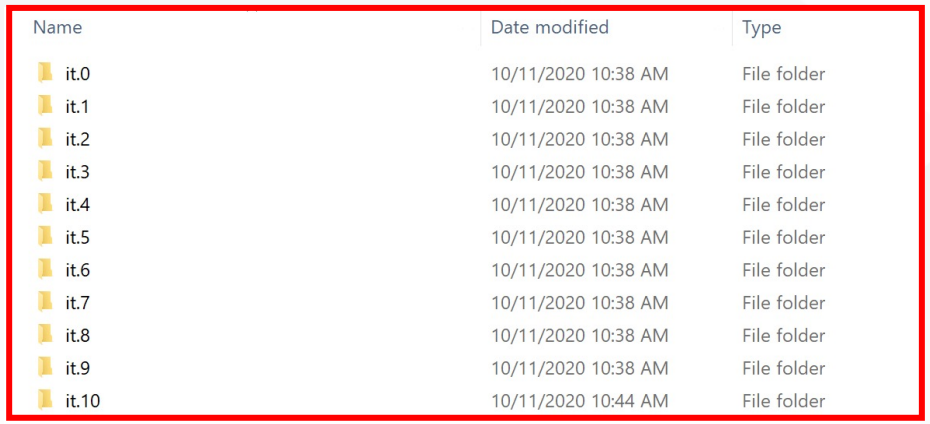
Overview of the output folder

Two Groups of Output Files

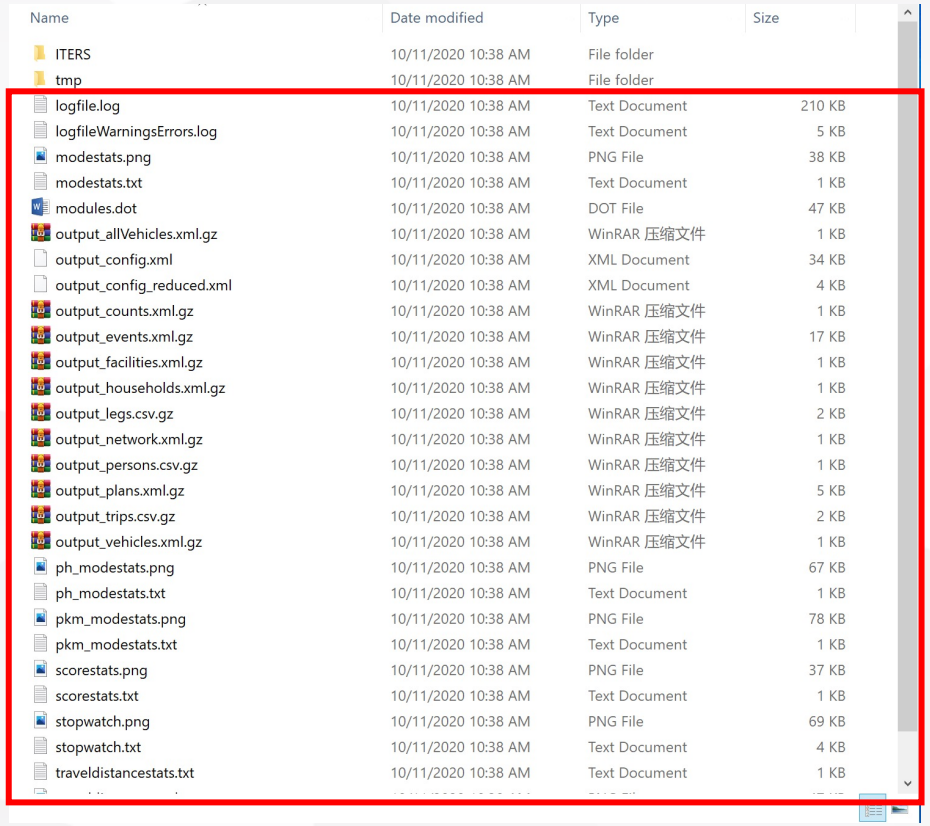
Path







(An example of output files for each iteration in the simulation)

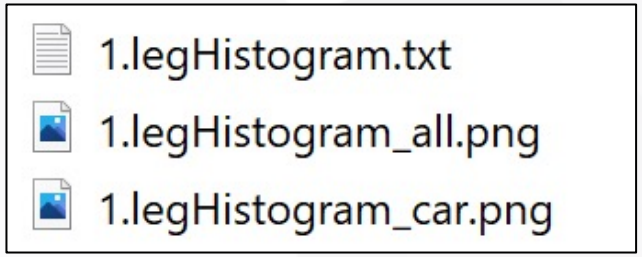


(An example of output files for the whole simulation)

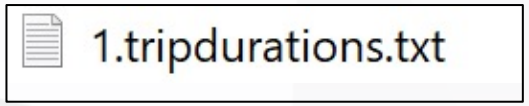
Output files stored in the iteration-specific folders

The following output files are generated for each iteration:

- **Leg Histogram**



- **Trip Durations**

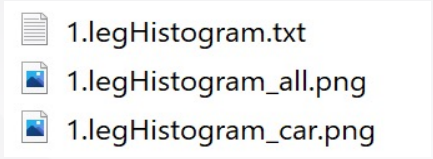


The following output files are generated for Specific iterations (every 10 iterations by default) :

- **Link States**

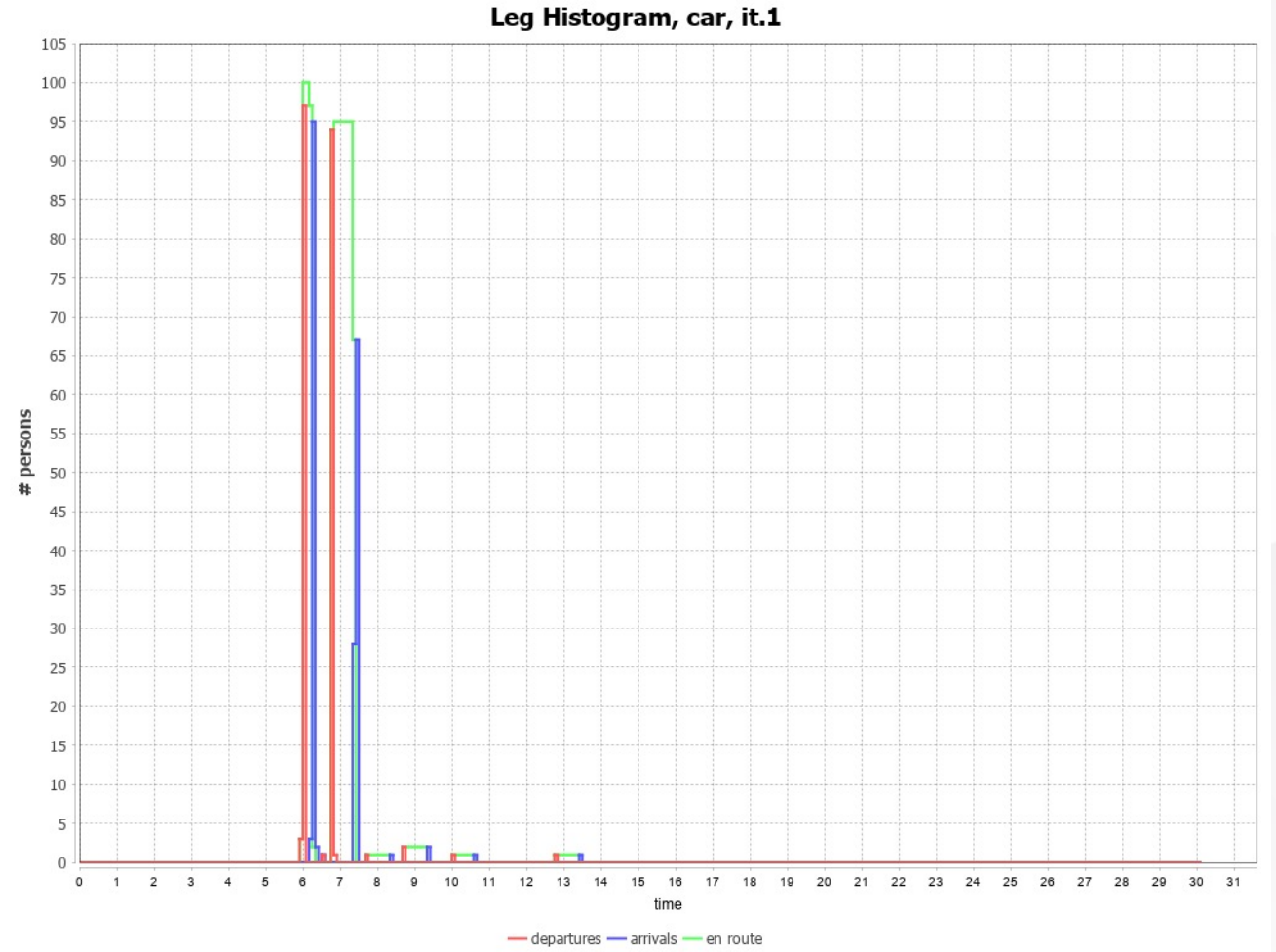


Output files stored in the iteration-specific folders



- **Leg Histogram**

  the number of agents arriving, departing or en route, per time unit



Output files stored in the iteration-specific folders

- **Trip Durations**

  the trip patterns for each trip type

1.tripdurations.txt

pattern	0+	5+	10+	15+	20+	25+	30+	35+	40+	45+	50+	55+	60+
h---w	0	0	0	98	2	0	0	0	0	0	0	0	0
w---h	0	0	0	0	0	0	0	100	0	0	0	0	0
w---w	1	0	0	0	0	0	0	0	0	0	0	0	0

average trip duration: 1678.1044776119404 seconds = 00:27:58

Output files stored in the iteration-specific folders



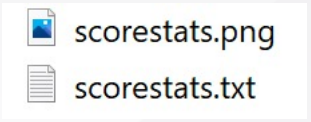
- **Link States**

  hourly count values and travel times of each road link (link states on a time bin level)

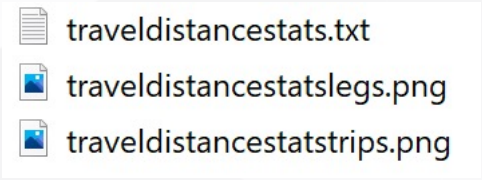
LINK	ORIG_ID	FROM	TO	LENGTH	FREESPEED	CAPACITY	HRS0-1min	HRS0-1avg	HRS0-1max	HRS1-2min	HRS1-2avg	HRS1-2max	HRS
1	1	2	10000.0	27.78	36000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 3.0 3.0	3.0 97.0	97.0	97.0
2	2	3	10000.0	27.78	3600.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0
3	2	4	10000.0	27.78	3600.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 10.0	10.8	11.0
4	2	5	10000.0	27.78	3600.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 4.0 9.6	15.0	0.0 0.0
5	2	6	10000.0	27.78	3600.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 8.0 8.6	9.0 0.0	0.0 0.0
6	2	7	10000.0	27.78	3600.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 38.0	47.0	56.0
7	2	8	10000.0	27.78	3600.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 6.0 8.8	17.0	0.0 0.0
8	2	9	10000.0	27.78	3600.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 5.0 7.4	12.0	0.0 0.0
9	2	10	10000.0	27.78	3600.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 4.0 7.8	15.0	0.0 0.0
10	2	11	10000.0	27.78	3600.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0
11	3	12	5000.0	27.78	1000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0
12	4	12	5000.0	27.78	1000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 10.0	10.8	11.0
13	5	12	5000.0	27.78	1000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 4.0 9.6	15.0	0.0 0.0
14	6	12	5000.0	27.78	1000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 8.0 8.6	9.0 0.0	0.0 0.0
15	7	12	5000.0	27.78	1000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 38.0	47.0	56.0
16	8	12	5000.0	27.78	1000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 6.0 8.8	17.0	0.0 0.0
17	9	12	5000.0	27.78	1000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 5.0 7.4	12.0	0.0 0.0
18	10	12	5000.0	27.78	1000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 4.0 7.8	15.0	0.0 0.0
19	11	12	5000.0	27.78	1000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0
20	12	13	10000.0	27.78	36000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 95.0	95.0	95.0
21	13	14	10000.0	27.78	36000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 95.0	95.0	95.0
22	14	15	35000.0	27.78	36000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 95.0	95.
23	15	1	10000.0	27.78	36000.0	0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 0.0 0.0	0.0 95.0	95.

Output files related to the whole simulation

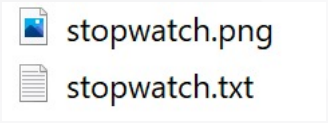
- **Score Statistics**



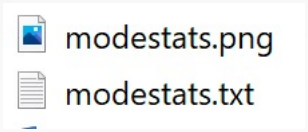
- **Travel Distance Statistics**



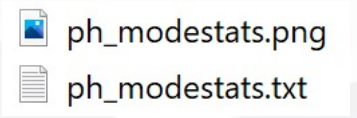
- **Stopwatch**



- **Mode statistics**



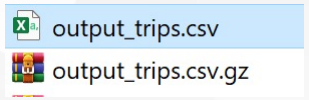
- **Passenger hours travelled per Mode**



- **Passenger kilometers travelled per Mode**



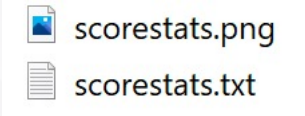
- **Trip information**



Output files related to the whole simulation

- **Score Statistics**

  the score of each iteration



C:\Users\daxiong\Desktop\matsim-12.0\examples\equil\output\scorestats.txt - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

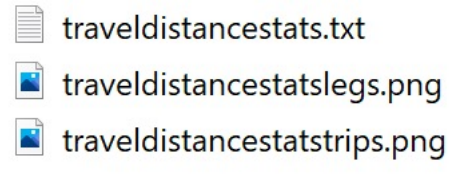
scorestats.txt

	ITERATION	avg. EXECUTED	avg. WORST	avg. AVG	avg. BEST
2	0	64.75686659291274	64.75686659291274	64.75686659291274	64.75686659291274
3	1	64.84180132563583	64.78366379257605	64.81273255910591	64.84180132563583
4	2	64.89143519950841	64.81165097216301	64.8538108966959	64.89199659582492
5	3	64.94635309581098	64.83660060721047	64.8953234518048	64.94691451976993
6	4	64.98358693977852	64.85666178537318	64.92565191373903	64.98376557505517
7	5	65.00593897973506	64.86984481213798	64.94610782801256	65.01006503556525
8	6	65.03313721370051	64.88314564317827	64.9646559908102	65.03313721370051
9	7	65.0434418130085	64.89177400633979	64.97561838443465	65.0442485132065
10	8	65.05962366496311	64.89351110119436	64.98594634490536	65.06089960576581
11	9	65.0833471487777	64.91171106319132	65.00921443880488	65.08498042435058
12	10	65.0866398879971	64.91592263882357	65.01554492675089	65.09045249606102

Output files related to the whole simulation

- **Travel Distance Statistics**

  the average travel distance of all trips in each iteration



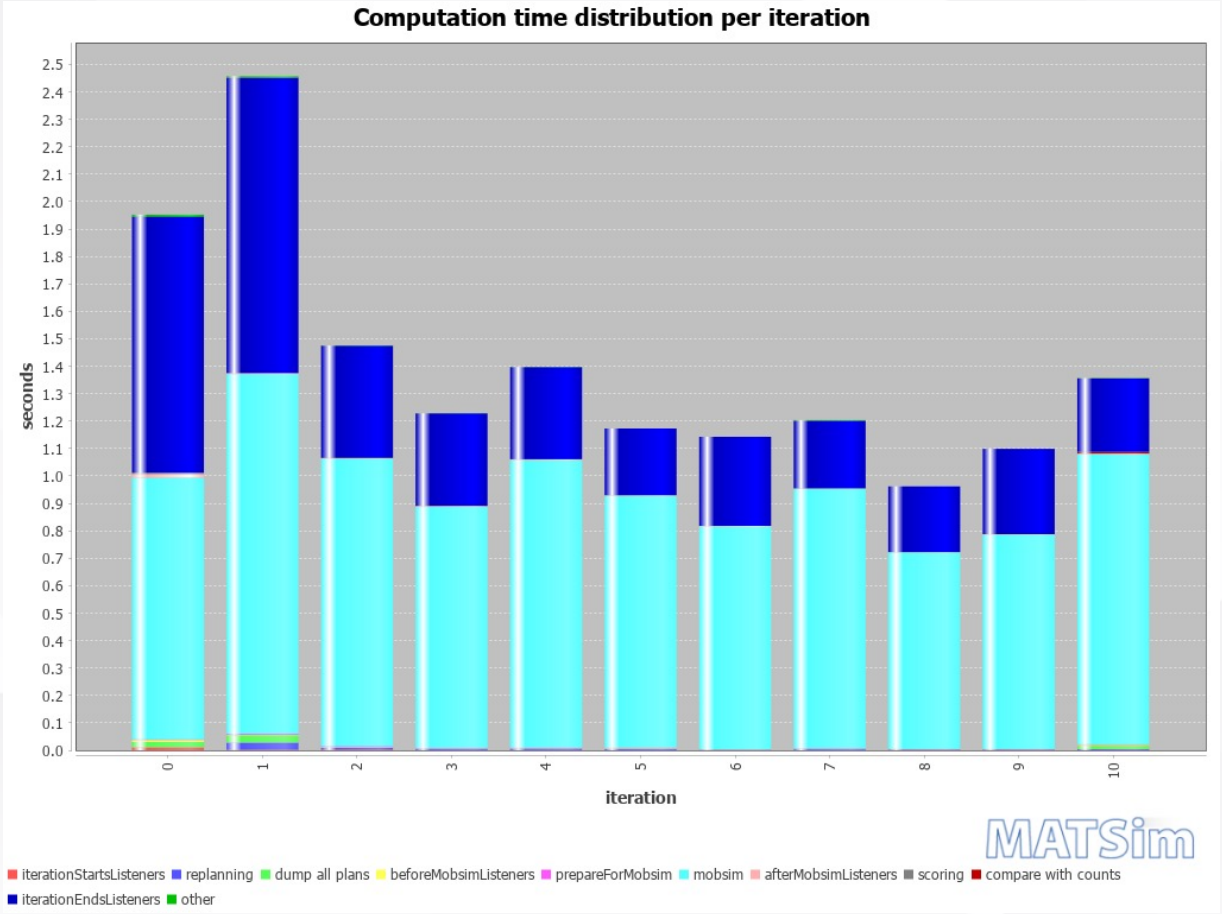
ITERATION	avg. Average Leg distance	avg. Average Trip distance
0	44776.119402985074	44776.119402985074
1	44776.119402985074	44776.119402985074
2	44776.119402985074	44776.119402985074
3	44776.119402985074	44776.119402985074
4	44776.119402985074	44776.119402985074
5	44776.119402985074	44776.119402985074
6	44776.119402985074	44776.119402985074
7	44776.119402985074	44776.119402985074
8	44776.119402985074	44776.119402985074
9	44776.119402985074	44776.119402985074
10	44776.119402985074	44776.119402985074

Output files related to the whole simulation

stopwatch.png
stopwatch.txt

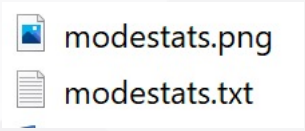
- **Stopwatch**

the computation time for *some typical actions* for each iteration



Output files related to the whole simulation

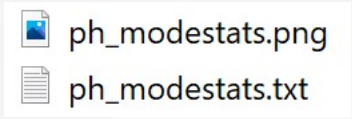
- **Mode statistics**
  the percentage of each mode used per iteration



Iteration	car	pt	walk
0	1.0	0.0	0.0
1	1.0	0.0	0.0
2	1.0	0.0	0.0
3	1.0	0.0	0.0
4	1.0	0.0	0.0
5	1.0	0.0	0.0
6	1.0	0.0	0.0
7	1.0	0.0	0.0
8	1.0	0.0	0.0
9	1.0	0.0	0.0
10	1.0	0.0	0.0

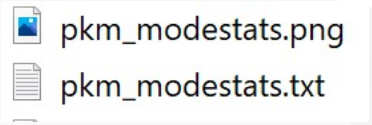
Output files related to the whole simulation

- **Passenger hours travelled per Mode**



the *total travel time* of all individuals using different modes in each iteration

- **Passenger kilometers travelled per Mode**



the *total travel distance* of all individuals using different modes in each iteration

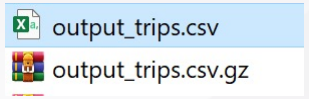
Iteration	car_travel	car_wait	stageActivity_travel	stageActivity_wait
0	94.64	0.0 0.0 0.0		
1	93.69416666666666	0.0 0.0 0.0		
2	93.13861111111112	0.0 0.0 0.0		
3	92.52222222222223	0.0 0.0 0.0		
4	92.07805555555555	0.0 0.0 0.0		
5	91.74194444444444	0.0 0.0 0.0		
6	91.52166666666666	0.0 0.0 0.0		
7	91.40694444444445	0.0 0.0 0.0		
8	91.21972222222222	0.0 0.0 0.0		
9	90.94777777777777	0.0 0.0 0.0		
10	90.91361111111111	0.0 0.0 0.0		

Iteration	car
0	9000.0
1	9000.0
2	9000.0
3	9000.0
4	9000.0
5	9000.0
6	9000.0
7	9000.0
8	9000.0
9	9000.0
10	9000.0

Output files related to the whole simulation

- **Trip information**

  detailed trip information of each person (travel patterns)



person	trip_number	trip_id	dep_time	trav_time	wait_time	traveled_distance	euclidean_distance	longest_distance_mode	modes	start_activity_type	end_activity_type	start_facility_id	start_link	start_x	start_y	end_facility_id	end_link	end_x	end_y	first_pt_boarding_stop	last_pt_
1	1	1_1	6:00:00	0:17:02	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
1	2	1_2	6:27:02	0:00:00	0:00:00	0	0		car	w	w	null	20	4950	-150	null	20	4950	-150		
1	3	1_3	9:57:02	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
2	1	2_1	5:59:00	0:15:00	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
2	2	2_2	8:44:00	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
3	1	3_1	5:58:30	0:15:00	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
3	2	3_2	7:43:30	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
4	1	4_1	5:59:30	0:15:00	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
4	2	4_2	8:44:30	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
5	1	5_1	6:00:00	0:15:39	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
5	2	5_2	12:45:39	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
6	1	6_1	6:00:00	0:15:08	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
6	2	6_2	6:45:08	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
7	1	7_1	6:00:00	0:15:39	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
7	2	7_2	6:45:39	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
8	1	8_1	6:00:00	0:15:25	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
8	2	8_2	6:45:25	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
9	1	9_1	6:00:00	0:15:07	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
9	2	9_2	6:45:07	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
10	1	10_1	6:00:00	0:15:35	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
10	2	10_2	6:45:35	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
11	1	11_1	6:00:00	0:15:39	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
11	2	11_2	6:45:39	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
12	1	12_1	6:00:00	0:16:55	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
12	2	12_2	6:46:55	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
13	1	13_1	6:00:00	0:15:14	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
13	2	13_2	6:45:14	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
14	1	14_1	6:00:00	0:15:32	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
14	2	14_2	6:45:32	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
15	1	15_1	6:00:00	0:15:28	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
15	2	15_2	6:45:28	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
16	1	16_1	6:00:00	0:16:51	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
16	2	16_2	6:46:51	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
17	1	17_1	6:00:00	0:15:28	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
17	2	17_2	6:45:28	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
18	1	18_1	6:00:00	0:16:47	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
18	2	18_2	6:46:47	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		
19	1	19_1	6:00:00	0:15:35	0:00:00	25000	19950	car	car	h	w	null	1	-15000	0	null	20	4950	-150		
19	2	19_2	6:45:35	0:39:00	0:00:00	65000	20000	car	car	w	h	null	20	4950	-150	null	1	-15050	-150		

Questions

If you have any technical problem with MATSim or our first 3 tutorials, you may turn to the Tutors for help.

- Ruichen MA (Email: ruichen.ma@connect.polyu.hk)

You may wish to copy Dr. Zhuge in when you send the Tutors an email.

- Chengxiang ZHUGE: chengxiang.zhuge@polyu.edu.hk